

TrackDel

Supply Chain Delivery Performance Dashboard

P R O J E C T R E P O R T

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Technologies

Python | MySQL | Power BI

TABLE OF CONTENTS

S.NO.	TOPIC	PAGE NO.
1	Introduction 1.1 Introduction 1.2 Executive Summary	1
2	Project Overview 2.1 Overall Description 2.2 Business Objective	2
3	Dataset Description 3.1 Dataset Information 3.2 Important Attributes 3.3 Purpose of Dataset	3
4	Data Cleaning and Preprocessing 4.1 Dataset Import 4.2 Missing Value Analysis 4.3 Column Removal 4.4 Missing Value Handling 4.5 Duplicate Record Check 4.6 Duplicate Removal 4.7 Date Conversion 4.8 Feature Engineering 4.9 Feature Verification 4.10 Export Cleaned Dataset	4
5	Exploratory Data Analysis (EDA) 5.1 Delivery Status Analysis 5.2 Late Delivery Risk Analysis 5.3 Shipping Mode Analysis 5.4 Order Status Analysis 5.5 Logistics Performance Analysis	7

6	Key Performance Indicators (KPIs) 6.1 Delivery Performance KPIs 6.2 Supplier Performance KPIs 6.3 Order Status KPIs 6.4 Logistics Efficiency KPIs	9
7	Visualizations Used 7.1 Delivery Delays Dashboard 7.2 Supplier Performance & Shipping Dashboard 7.3 Order Status Dashboard 7.4 Logistics Efficiency Dashboard	10
8	Key Insights 8.1 Delivery Performance Insights 8.2 Shipping Insights 8.3 Order Status Insights 8.4 Supplier Performance Insights	14
9	Recommendations 9.1 Delivery Performance Recommendations 9.2 Shipping Optimization Recommendations 9.3 Order Management Recommendations 9.4 Business Performance Recommendations	15
10	Conclusion	16
11	Tools and Technologies Used	17
12	References	18

CHAPTER 1

INTRODUCTION

1.1 Introduction

Supply chain management plays a vital role in ensuring that products are delivered to customers efficiently, on time, and at minimum operational cost. With the rapid growth of e-commerce and global business operations, organizations generate large volumes of logistics and delivery data every day. Analyzing this data helps businesses identify delivery delays, monitor shipping performance, evaluate supplier efficiency, and improve overall operational effectiveness.

TrackDel is a data analytics project developed to analyze supply chain delivery performance using historical order and shipment data. The project utilizes Python for data cleaning and preprocessing, MySQL for data storage and query execution, and Power BI for creating interactive dashboards. The dashboards provide meaningful insights into delivery performance, supplier performance, shipping efficiency, order status, and logistics efficiency.

By transforming raw supply chain data into interactive visual reports, the project supports data-driven decision-making and helps organizations improve customer satisfaction, reduce delivery delays, and optimize logistics operations.

1.2 Executive Summary

TrackDel is a Supply Chain Delivery Performance Dashboard developed using Python, MySQL, and Power BI to analyze and visualize supply chain operations. The project focuses on identifying delivery delays, evaluating supplier performance, monitoring shipping performance, analyzing order status, and measuring logistics efficiency using historical supply chain data.

The dataset contains **180,519 records and 53 columns** related to customers, products, orders, sales, shipping, and delivery. After cleaning and preprocessing the data, interactive dashboards were created to monitor key performance indicators (KPIs) and generate actionable business insights.

The project helps identify delayed deliveries, evaluate shipping mode performance, analyze sales and profit trends, monitor order completion, and improve logistics efficiency. Overall, TrackDel demonstrates how business intelligence tools can transform raw operational data into meaningful insights for effective supply chain management.

CHAPTER 2

PROJECT OVERVIEW

2.1 Overall Description

TrackDel is a Supply Chain Delivery Performance Dashboard developed to analyze historical supply chain data and monitor delivery performance. The project focuses on identifying delivery delays, evaluating supplier performance, analyzing shipping efficiency, monitoring order status, and measuring logistics performance through interactive dashboards.

Python was used for data cleaning and preprocessing, MySQL for storing and managing the cleaned dataset, and Power BI for developing interactive dashboards and visualizations. The project is based on the **DataCo Supply Chain Dataset**, containing **180,519 records** and **53 attributes**. The developed dashboards provide key performance indicators (KPIs) and business insights that support data-driven decision-making and help improve overall supply chain efficiency.

2.2 Business Objective

The primary objective of the TrackDel project is to analyze supply chain delivery performance and provide meaningful insights through interactive dashboards. By using Python for data preprocessing, MySQL for data management, and Power BI for visualization, the project enables effective monitoring of key supply chain activities and supports better business decision-making.

The main objectives of the project are:

1. Analyze delivery performance across different regions.
2. Identify delayed deliveries and compare them with on-time deliveries.
3. Evaluate the performance of different shipping modes.
4. Analyze supplier and product category performance using sales and profit metrics.
5. Monitor order status distribution, including completed, pending, cancelled, and fraud review orders.
6. Measure logistics efficiency using key performance indicators (KPIs).
7. Develop interactive dashboards for monitoring supply chain operations.
8. Generate actionable business insights to improve delivery performance and reduce delays.
9. Support data-driven decision-making through clear visualizations and performance metrics.
10. Identify opportunities for improving logistics operations, shipping efficiency, and overall supply chain management.

CHAPTER 3

DATASET DESCRIPTION

3.1 Dataset Information

- Dataset Name: DataCo Supply Chain Dataset
- Source: Kaggle
- Total Records: 180,519
- Total Columns: 53

3.2 Important Attributes

Table 3.1: Important Dataset Attributes

Column Name	Description
Order ID	Unique order identifier
Order Status	Current order status
Delivery Status	Delivery performance status
Shipping Mode	Mode of shipment
Category Name	Product category
Sales	Revenue generated
Order Profit Per Order	Profit generated
Delay_Days	Delivery delay in days
Order Region	Delivery region
DateOrders	Order date

3.3 Purpose of Dataset

The dataset was used to analyze:

- Delivery performance
- Shipping efficiency
- Order status distribution
- Supplier performance
- Logistics operations
- Regional delivery delays

CHAPTER 4

DATA CLEANING AND PREPROCESSING

4.1 Dataset Import

The supply chain dataset was imported into Python using the Pandas library for preprocessing.

Code:

```
import pandas as pd
```

```
df = pd.read_csv("DataCoSupplyChainDataset.csv", encoding="ISO-8859-1")  
df.shape
```

Observation:

The dataset was successfully loaded into Python and its dimensions were verified.

4.2 Missing Value Analysis

Missing values were identified to determine which columns required preprocessing.

Code:

```
df.isnull().sum()
```

Observation:

- Customer Lname contained 8 missing values.
- Customer Zipcode contained 3 missing values.
- Order Zipcode contained a large number of missing values.

4.3 Column Removal

The unnecessary column was removed from the dataset.

Code:

```
df.drop(columns=['Product Description'], inplace=True)  
df.shape
```

Observation:

The **Product Description** column was completely empty and was removed from the dataset.

4.4 Missing Value Handling

Missing values in important columns were filled to improve data quality.

Code:

```
df['Customer Lname'].fillna('Unknown', inplace=True)
df['Customer Zipcode'].fillna(0, inplace=True)
```

Observation:

- Missing values in **Customer Lname** were replaced with "**Unknown**".
- Missing values in **Customer Zipcode** were replaced with **0**.

4.5 Duplicate Record Check

Duplicate records were checked before further analysis.

Code:

```
df.duplicated().sum()
```

Observation:

No duplicate records were found in the dataset.

4.6 Duplicate Removal

Duplicate records were removed if present.

Code:

```
df.drop_duplicates(inplace=True)
df.shape
```

Observation:

The dataset was verified to be free of duplicate records.

4.7 Date Conversion

Date columns were converted into datetime format for analysis.

Code:

```
df['order date (DateOrders)'] = pd.to_datetime(df['order date (DateOrders)'])
df['shipping date (DateOrders)'] = pd.to_datetime(df['shipping date (DateOrders)'])
```

Observation:

Order date and shipping date columns were successfully converted into datetime format.

4.8 Feature Engineering

A new feature was created to calculate delivery delay.

Code:

```
df['Delay_Days'] = (  
    df['Days for shipping (real)']  
    - df['Days for shipment (scheduled)']  
)
```

Observation:

A new column **Delay_Days** was created by subtracting scheduled shipping days from actual shipping days.

4.9 Feature Verification

The newly created feature was validated using descriptive statistics.

Code:

```
df['Delay_Days'].describe()
```

Observation:

The Delay_Days column was successfully generated and verified.

4.10 Export Cleaned Dataset

The cleaned dataset was exported for SQL integration and Power BI dashboard development.

Code:

```
df.to_csv("TrackDel_Cleaned.csv", index=False)
```

Observation:

The cleaned dataset was exported as **TrackDel_Cleaned.csv** and used for MySQL and Power BI.

CHAPTER 5

EXPLORATORY DATA ANALYSIS (EDA)

5.1 Delivery Status Analysis

Table 5.1: Delivery Status Analysis

Delivery Status	Orders
Late Delivery	98,977
Advance Shipping	41,592
Shipping On Time	32,196
Shipping Canceled	7,754

Observation:

Late delivery is the most common delivery status, indicating significant delivery performance challenges.

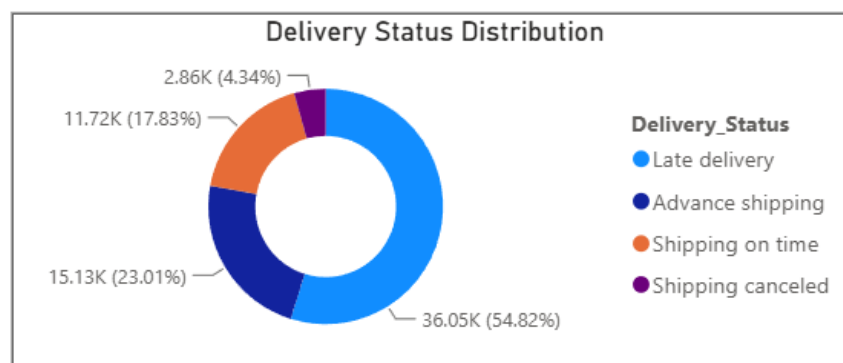


Figure 5.1: Delivery Status Distribution (Bar Chart)

5.2 Late Delivery Risk Analysis

Table 5.2: Late Delivery Risk Analysis

Status	Orders
Late Delivery Risk	98,977
On-Time Delivery	81,542

Observation:

More than half of the deliveries were associated with late delivery risk.

5.3 Shipping Mode Analysis

Table 5.3: Shipping Mode Analysis

Shipping Mode	Orders
Standard Class	107,752
Second Class	35,216
First Class	27,814
Same Day	9,737

Observation:

Standard Class is the most frequently used shipping mode.

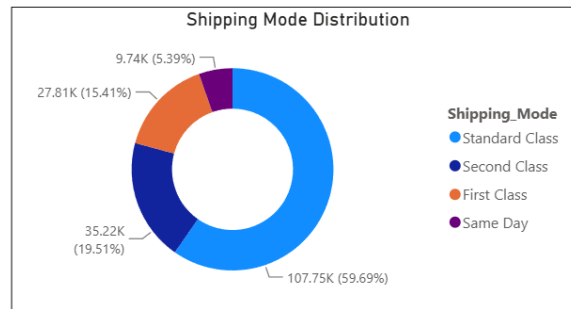


Figure 5.2: Shipping Mode Distribution (Pie Chart)

5.4 Order Status Analysis

Table 5.4: Order Status Analysis

Order Status	Orders
COMPLETE	59,491
PENDING_PAYMENT	39,832
PROCESSING	21,902
PENDING	20,227
CLOSED	19,616
ON_HOLD	9,804
SUSPECTED_FRAUD	4,062
CANCELED	3,692
PAYMENT_REVIEW	1,893

Observation:

Completed orders represent the largest share of transactions.

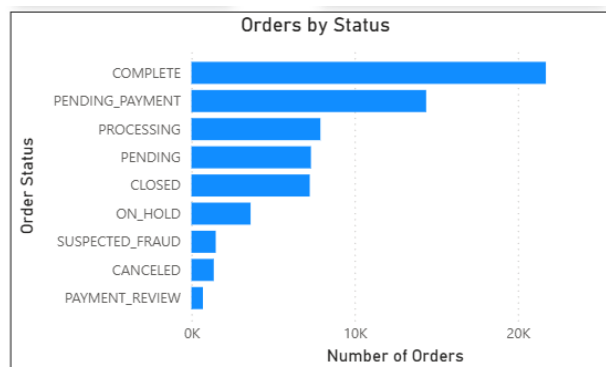


Figure 5.3: Orders by Status (Horizontal Bar Chart)

5.5 Logistics Performance Analysis

The average actual shipping time was higher than the average scheduled shipping time.

Observation:

The difference between actual and scheduled shipping days indicates opportunities for improving logistics efficiency.

CHAPTER 6

KEY PERFORMANCE INDICATORS (KPIs)

6.1 Delivery Performance KPIs

Table 6.1: Delivery Performance KPIs

KPI	Value
Total Orders	66K
On-Time Orders	12K
Late Orders	36K
Late Delivery %	54.82%
Average Delay Days	0.57

6.2 Supplier Performance KPIs

Table 6.2: Supplier Performance KPIs

KPI	Value
Total Sales	36.78M
Total Profit	3.97M
Average Profit per Order	21.97
Average Shipping Days	3.50
Most Used Shipping Mode	Standard Class

6.3 Order Status KPIs

Table 6.3: Order Status KPIs

KPI	Value
Completed Orders	29K
Pending & Processing Orders	33K
Cancelled Orders	1K
Fraud Review Orders	2K
Completion Rate %	44.05%

6.4 Logistics Efficiency KPIs

Table 6.4: Logistics Efficiency KPIs

KPI	Value
Average Scheduled Days	2.93
Shipping Efficiency %	83.82%
On-Time Shipping %	17.83%
Schedule Adherence %	42.67%
Most Delayed Shipping Mode	Second Class

CHAPTER 7

VISUALIZATIONS USED

7.1 Delivery Delays Dashboard

The Delivery Delays Dashboard was developed to analyze delayed deliveries and monitor delivery performance across different regions.

Visualizations Used:

- KPI Cards (Total Orders, On-Time Orders, Late Orders, Late Delivery %, Average Delay Days)
- Bar Chart (Late Deliveries by Region)
- Donut Chart (Delivery Status Distribution)
- Bar Chart (Average Delay Days by Shipping Mode)
- Region Filter
- Shipping Mode Filter

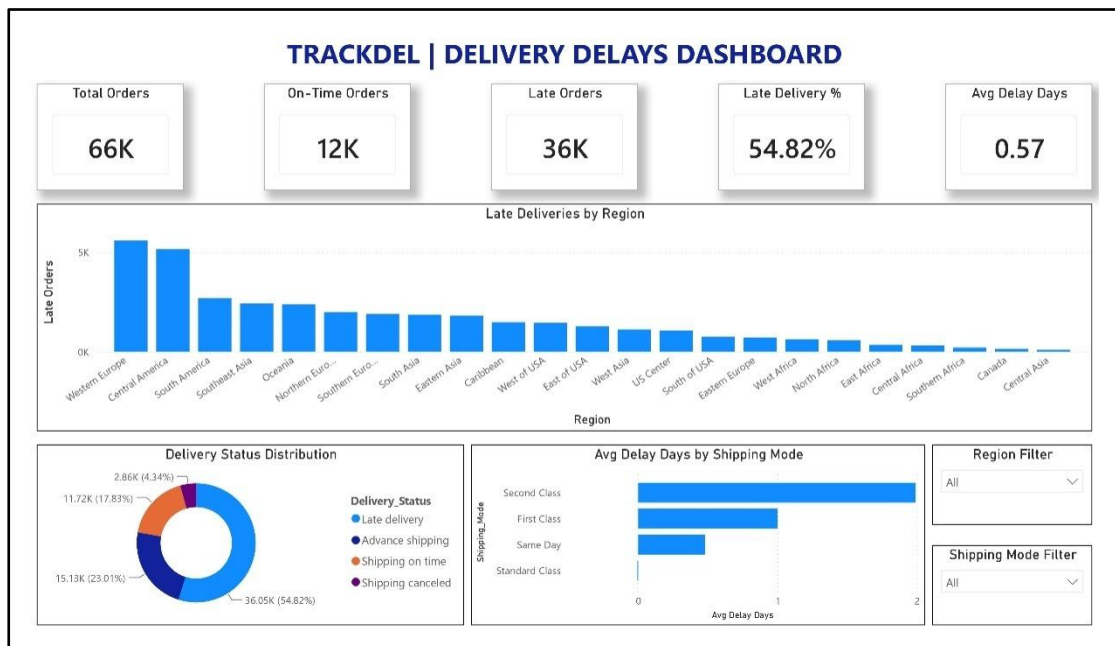


Figure 7.1: Delivery Delays Dashboard

7.2 Supplier Performance & Shipping Dashboard

This dashboard focuses on supplier performance, category-wise sales, profit analysis, and shipping utilization.

Visualizations Used:

- KPI Cards (Total Sales, Total Profit, Average Profit per Order, Average Shipping Days, Most Used Shipping Mode)
- Bar Chart (Top 10 Categories by Sales)
- Bar Chart (Top 10 Categories by Order Profit)
- Donut Chart (Shipping Mode Distribution)
- Region Filter
- Category Filter

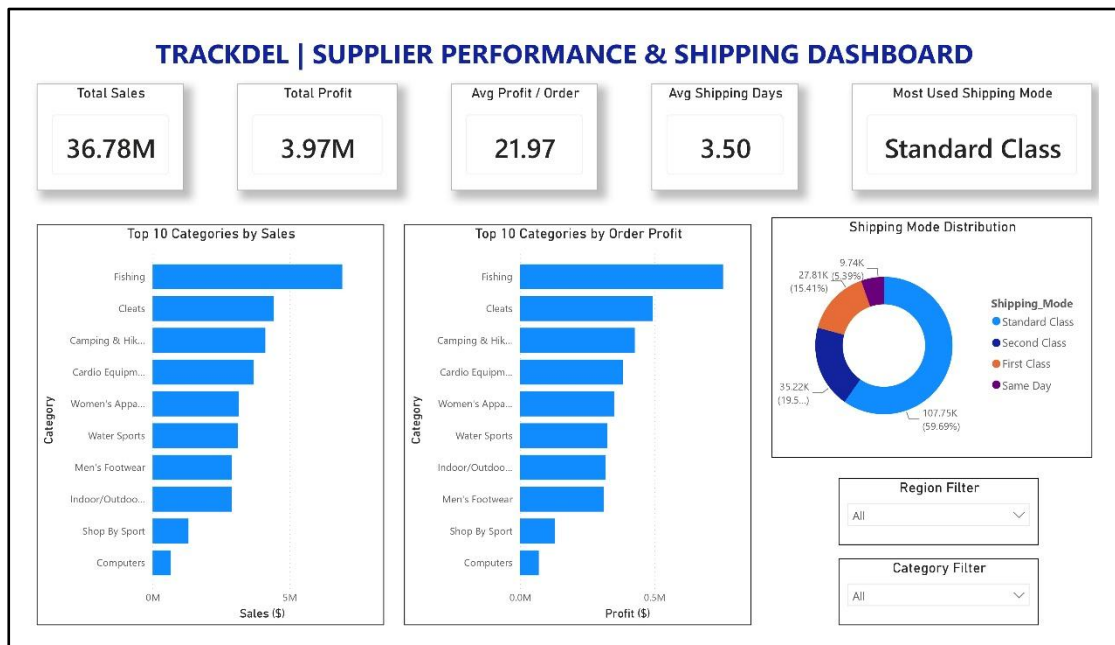


Figure 7.2: Supplier Performance & Shipping Dashboard

7.3 Order Status Dashboard

This dashboard was developed to monitor order completion, pending orders, cancelled orders, and fraud review orders.

Visualizations Used:

- KPI Cards (Completed Orders, Pending & Processing Orders, Cancelled Orders, Fraud Review Orders, Completion Rate %)
- Bar Chart (Orders by Status)
- Stacked Bar Chart (Order Status by Region)
- Donut Chart (Order Status Distribution)
- Region Filter
- Status Filter

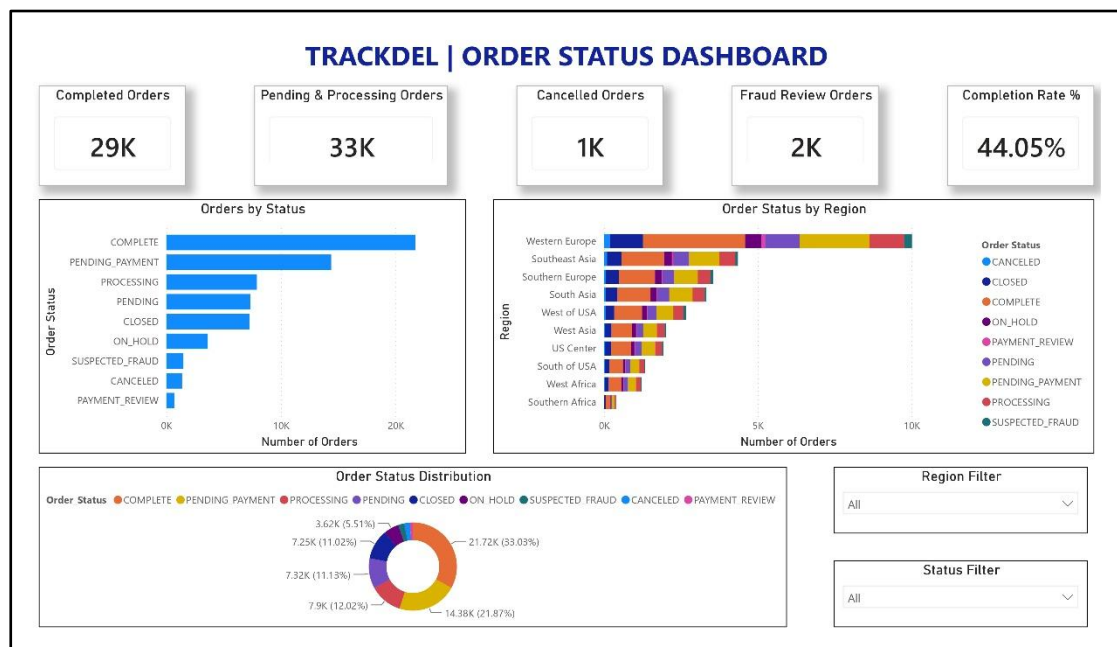


Figure 7.3: Order Status Dashboard

7.4 Logistics Efficiency Dashboard

This dashboard evaluates shipping efficiency and logistics performance.

Visualizations Used:

- KPI Cards (Avg Scheduled Days, Shipping Efficiency %, On-Time Shipping %, Schedule Adherence %, Most Delayed Shipping Mode)
- Bar Chart (Average Shipping Time by Shipping Mode)
- Bar Chart (Average Delay by Region)
- Donut Chart (Shipping Mode Efficiency)
- Clustered Column Chart (Actual vs Scheduled Shipping Days)
- Region Filter
- Shipping Mode Filter

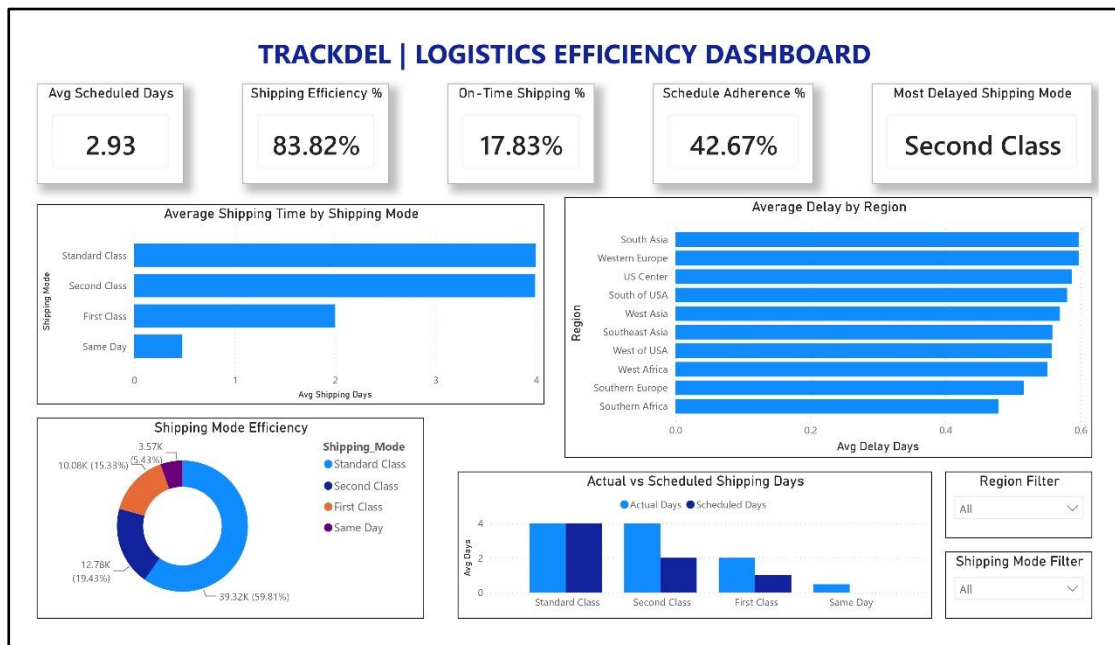


Figure 7.4: Logistics Efficiency Dashboard

CHAPTER 8

KEY INSIGHTS

8.1 Delivery Performance Insights

- More than half of the orders experienced late delivery.
- Late delivery percentage exceeded 50%, indicating delivery performance challenges.
- Certain regions showed significantly higher delay levels than others.

8.2 Shipping Insights

- Standard Class was the most frequently used shipping mode.
- Second Class showed the highest average delay among shipping modes.
- Actual shipping days were generally higher than scheduled shipping days.

8.3 Order Status Insights

- Completed orders represented the largest share of processed orders.
- A significant number of orders remained in pending and processing stages.
- Cancelled and fraud review orders represented a small portion of total orders.

8.4 Supplier Performance Insights

- A few product categories contributed the majority of sales and profit.
- Category performance varied significantly across the dataset.

CHAPTER 9

RECOMMENDATIONS

9.1 Delivery Performance Recommendations

1. Improve monitoring of delayed shipments in high-risk regions.
2. Implement proactive tracking mechanisms for delayed orders.
3. Reduce delivery bottlenecks through better route planning.

9.2 Shipping Optimization Recommendations

1. Review the performance of Second Class shipping mode.
2. Increase the use of efficient shipping methods for critical deliveries.
3. Regularly monitor actual shipping days against scheduled shipping days.

9.3 Order Management Recommendations

1. Reduce pending and processing orders through workflow optimization.
2. Monitor fraud review orders to minimize order processing delays.
3. Improve order fulfillment processes to increase completion rates.

9.4 Business Performance Recommendations

1. Focus on high-performing product categories to maximize profitability.
2. Analyze low-performing categories and improve their performance.
3. Use dashboard insights for data-driven decision-making.

CHAPTER 10

CONCLUSION

The TrackDel project successfully analyzed supply chain delivery performance using Python, MySQL, and Power BI.

The analysis revealed significant delivery delays, variations in shipping performance, and differences in order status distribution across regions. Interactive dashboards were developed to monitor delivery performance, supplier performance, order processing, and logistics efficiency.

The insights generated from this project can help organizations improve delivery operations, optimize shipping performance, reduce delays, and enhance overall supply chain efficiency.

TrackDel demonstrates how data analytics and business intelligence tools can be used to transform raw supply chain data into meaningful business insights and support informed decision-making.

CHAPTER 11

TOOLS AND TECHNOLOGIES USED

- Python (Pandas)
- MySQL
- Power BI
- Visual Studio Code
- Microsoft Excel

Purpose of Tools and Technologies Used

Table 11.1: Purpose of Tools and Technologies Used

Tool	Purpose
Python (Pandas)	Data Cleaning, Feature Engineering and EDA
MySQL	Data Storage and Query Execution
Power BI	Dashboard Development and Visualization
Visual Studio Code	Development Environment
Microsoft Excel	Dataset Verification and Inspection

CHAPTER 12

REFERENCES

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